

Royce Mathew

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Experience

Full-Stack Developer | TD Asset Management

January 2026 - Present

- Built portfolio analytics tools for the **Portfolio Analytics team**, enabling portfolio managers to monitor and act on trading data across **billions of dollars** in assets, with an emphasis on performance and accuracy.
- Engineered **guideline monitoring tools** for PMs to validate portfolios against **benchmark-relative, leverage ratio**, and security-filtering rules, including a view visualizing each security's contribution to a given rule.
- Developed start-of-day reports and a notification system for PMs and distribution lists to subscribe to updates based on continuous client feedback.

Cloud Developer | Royal Bank of Canada (RBC)

January 2024 - August 2024

- Developed an application using React, TypeScript, and Microsoft SQL that was utilized by **50+ developers** and compliance teams to track vulnerability scores for **15,000+ enterprise devices**.
- Integrated Azure Entra ID with ServiceNow for secure access control. Built scalable cloud infrastructure with Docker and OpenShift with automated horizontal scaling and load balancing configurations.
- Automated CI/CD pipelines (Jenkins, SonarQube, Checkmarx, Artifactory) and implemented AI-assisted unit tests, **reducing query time ~30%** through caching and SQL optimization.

Teaching Assistant | Ontario Tech University

September 2023 - April 2025

- Mentored **100+ students** across 3 courses on data structures, earning **90% positive feedback** in student evaluations.

Technical Operations Intern | Modern Deploy Technology

May 2023 - September 2023

- Built a document management application using Svelte, TypeScript, and Supabase authentication, enabling secure access for **200+ users** from different corporations. Implemented a PostgreSQL Database for cloud storage.

Projects

[rOyce.com/projects](https://royce.com/projects)

Kanjou - Collaborative Text Editor

Firebase • NextJs • React • Typescript

- Developed a real-time collaborative text editor with concurrent multi-user editing and conflict resolution, achieving **sub-100ms** latency via Firebase onSnapshot listeners.
- Implemented rich text editing with syntax highlighting for **20+ programming languages** using the Tiptap editor, and a custom middleware solution for Firebase adapter authentication in a serverless environment.

Midas - Trading Engine

Python • FastAPI • React • Docker • SQLite

- Built an algorithmic trading engine for **NQ/ES futures** that detects institutional price patterns (fair value gaps, liquidity sweeps, market structure shifts) across multiple timeframes, scoring and grading signals in real time.
- Engineered a backtesting system that replays historical data through the exact same signal pipeline used live, eliminating lookahead bias and ensuring backtest/live parity.

Education

University of Ontario Institute of Technology

September 2021 - December 2025

Bachelor of Science (Honours) in Computer Science - 3.89 GPA

- Dean's and President's Honors List** 2022 - 2025
- Relevant Courses:** Machine Learning, Systems Programming, Algorithms & Data Structures, Compilers

Skills

Languages: Python, Java, Rust, Go, JavaScript, SQL, C++, C#, C, Assembly, Lua

Frameworks & Tools: React, Next.js, Flask, Docker, Jenkins, Git, REST APIs, Svelte, Selenium, Bash

Cloud Platforms: Google Cloud, Microsoft Azure, AWS, Firebase, Red Hat OpenShift

DevOps & Security: HashiCorp Vault, SonarQube, Artifactory

Data & ML: PyTorch, TensorFlow, Scikit-Learn, Pandas, NumPy, PostgreSQL, MSSQL, DuckDB